NanoGT Match Report: Vitamin B12-unresponsive methylmalonic acidemia

Disease: Vitamin B12-unresponsive methylmalonic acidemia (ORPHA:27)
Primary gene: MMUT
Gene CDS: 2250 bp
Inheritance: Autosomal recessive
Target tissues scored: liver, CNS

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: MITOCHONDRIAL MATRIX ENZYME (MMUT): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.; Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints; Cell-autonomous protein across multiple tissues; high transduction coverage may be required.

Disease Mechanism Evidence

Molecular mechanism: loss of function
Mechanistic detail: Methylmalonyl-CoA mutase — nuclear-encoded mitochondrial MATRIX enzyme requiring post-translational import via intact N-terminal MTS
Gene-addition compatibility: conditional
Preferred modality class: liver aav with mts validation
Evidence level/status: direct / source_linked_needs_review
Evidence summary: MUT is a nuclear-encoded mitochondrial matrix enzyme. Nuclear AAV delivery is theoretically feasible but the N-terminal mitochondrial targeting sequence (MTS) must be intact in the therapeutic construct for correct post-translational import into the matrix. MTS functionality and import efficiency must be validated experimentally before vector precedent scores can be applied. Liver-directed AAV programs in development (e.g. NCT03721861) confirm the approach is viable but disease-specific construct validation is required.
Evidence source: OMIM MUT gene entry

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and integrating ex vivo HSC vector precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Endpoint readiness: liver/metabolic targets may have biochemical biomarkers, but biomarker correction must be linked to clinical benefit.
- Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
- Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	BMN 307	AAV5	7.7/10	High	phase2
2	DTX301	AAV8	7.1/10	Medium	phase2
3	Libmeldy	LV	6.7/10	Medium	approved
4	OAV101-IT	AAV9	6.7/10	Medium	approved
5	Zolgensma	AAV9	6.7/10	Medium	approved

Match #1: BMN 307

Precedent disease: Phenylketonuria
Vector: AAV5
Tissue target: liver
Composite score: 7.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.71**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 2250bp / cargo 4700bp (48% utilized)

- Vector tropism plus precedent target match: liver
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: amino_acid_metabolism
- Disease mechanism: loss of function — Methylmalonyl-CoA mutase — nuclear-encoded mitochondrial MATRIX enzyme requiring post-translational import via intact N-terminal MTS
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MUT is a nuclear-encoded mitochondrial matrix enzyme. Nuclear AAV delivery is theoretically feasible but the N-terminal mitochondrial targeting sequence (MTS) must be intact in the therapeutic construct for correct post-translational import into the matrix. MTS functionality and import efficiency must be validated experimentally before vector precedent scores can be applied. Liver-directed AAV programs in development (e.g. NCT03721861) confirm the approach is viable but disease-specific construct validation is required.
- Mechanism source: OMIM MUT gene entry (https://omim.org/entry/609058)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: CONDITIONAL — MMUT is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import efficiency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (MMUT): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: DTX301

Precedent disease: Ornithine transcarbamylase deficiency
Vector: AAV8
Tissue target: liver
Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.10**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 2250bp / cargo 4700bp (48% utilized)
- Vector tropism plus precedent target match: liver
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Pathway match: urea_cycle
- Disease mechanism: loss of function — Methylmalonyl-CoA mutase — nuclear-encoded mitochondrial MATRIX enzyme requiring post-translational import via intact N-terminal MTS
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MUT is a nuclear-encoded mitochondrial matrix enzyme. Nuclear AAV delivery is theoretically feasible but the N-terminal mitochondrial targeting sequence (MTS) must be intact in the therapeutic construct for correct post-translational import into the matrix. MTS functionality and import efficiency must be validated experimentally before vector prece-

dent scores can be applied. Liver-directed AAV programs in development (e.g. NCT03721861) confirm the approach is viable but disease-specific construct validation is required.
- Mechanism source: OMIM MUT gene entry (https://omim.org/entry/609058)
- Approval status: phase2
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: CONDITIONAL — MMUT is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import efficiency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (MMUT): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: Libmeldy

Precedent disease: Metachromatic leukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**6.71**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 2250bp / cargo 8000bp (28% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (amino_acid_metabolism vs leukodystrophy)
- Disease mechanism: loss of function — Methylmalonyl-CoA mutase — nuclear-encoded mitochondrial MATRIX enzyme requiring post-translational import via intact N-terminal MTS
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MUT is a nuclear-encoded mitochondrial matrix enzyme. Nuclear AAV delivery is theoretically feasible but the N-terminal mitochondrial targeting sequence (MTS) must be intact in the therapeutic construct for correct post-translational import into the matrix. MTS functionality and import efficiency must be validated experimentally before vector precedent scores can be applied. Liver-directed AAV programs in development (e.g. NCT03721861) confirm the approach is viable but disease-specific construct validation is required.
- Mechanism source: OMIM MUT gene entry (https://omim.org/entry/609058)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or

more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: CONDITIONAL — MMUT is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import efficiency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (MMUT): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, liver
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable

Match #4: OAV101-IT

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/spinal cord
Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage

Dimension	Score	Max	What it measures
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**6.71**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 2250bp / cargo 4700bp (48% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (amino_acid_metabolism vs motor_neuron)
- Disease mechanism: loss of function — Methylmalonyl-CoA mutase — nuclear-encoded mitochondrial MATRIX enzyme requiring post-translational import via intact N-terminal MTS
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MUT is a nuclear-encoded mitochondrial matrix enzyme. Nuclear AAV delivery is theoretically feasible but the N-terminal mitochondrial targeting sequence (MTS) must be intact in the therapeutic construct for correct post-translational import into the matrix. MTS functionality and import efficiency must be validated experimentally before vector precedent scores can be applied. Liver-directed AAV programs in development (e.g. NCT03721861) confirm the approach is viable but disease-specific construct validation is required.
- Mechanism source: OMIM MUT gene entry (https://omim.org/entry/609058)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: CONDITIONAL — MMUT is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import effi-

ciency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (MMUT): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: Zolgensma

Precedent disease: Spinal Muscular Atrophy
Vector: AAV9
Tissue target: CNS/motor neuron
Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist

Dimension	Score	Max	What it measures
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	0.50	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**6.71**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 2250bp / cargo 4700bp (48% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (amino_acid_metabolism vs motor_neuron)
- Disease mechanism: loss of function — Methylmalonyl-CoA mutase — nuclear-encoded mitochondrial MATRIX enzyme requiring post-translational import via intact N-terminal MTS
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: MUT is a nuclear-encoded mitochondrial matrix enzyme. Nuclear AAV delivery is theoretically feasible but the N-terminal mitochondrial targeting sequence (MTS) must be intact in the therapeutic construct for correct post-translational import into the matrix. MTS functionality and import efficiency must be validated experimentally before vector precedent scores can be applied. Liver-directed AAV programs in development (e.g. NCT03721861) confirm the approach is viable but disease-specific construct validation is required.
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- ORGANELLE TARGETING: CONDITIONAL — MMUT is a nuclear-encoded mitochondrial matrix protein. Nuclear AAV delivery is theoretically feasible (the gene is in the nuclear genome) but the N-terminal mitochondrial targeting sequence (MTS) must be preserved intact in the therapeutic construct for correct post-translational import into the mitochondrial matrix. MTS functionality is not validated by any other dimension in this framework. Confirm import efficiency with disease-specific in vitro/in vivo data before treating vector precedent scores as directly transferable.

Manual Review Flags

- MITOCHONDRIAL MATRIX ENZYME (MMUT): nuclear-encoded but the protein must be imported into the mitochondrial matrix post-translation via its N-terminal mitochondrial targeting sequence (MTS). Nuclear AAV delivery is theoretically feasible, but the therapeutic construct must preserve the intact MTS. MTS functionality is not captured by any other scoring dimension — this is an additional disease-specific development step requiring experimental validation.

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone